Muhammad Zaid

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Lahore, Pakistan

Professional Summary

Computer science graduate experienced in building software in Python and C++, including web publishing automation (Notes Publisher), multi-threaded graphics (Ray Tracing) and game development (Memory Maze). GPA: 3.7/4.0.

Skills

- **Languages:** C, C++, Python, JavaScript (JS), LaTeX, Typst, YAML
- **Tools:** SFML, build system (CMake), Version Control (Git)
- **Concepts:** Object Oriented Programming (OOP), Data Structure and algorithm design (DSA), Software Design Patterns (Observer, Strategy)
- **Others:** Abstract and critical thinking, analytical problem-solving, technical writing

Projects

Notes Publisher https://github.com/ChuzaWick420/notes_publisher

- Designed an automated personal knowledge-base pipeline that converts Markdown notes into a published static website using MkDocs, YAML, Bash, Python, and LaTeX.
- Designed an automated backup workflow using Google Drive, GitHub, and GitLab to provide redundant storage and reduce risk of data loss from hardware or power failures.

Memory Maze - FYP https://github.com/ChuzaWick420/FYP_AI_Based_Puzzle_Game

- Built a puzzle and layered game engine in Python, applying the Observer and Strategy design patterns to decouple game logic from the UI (made with Pygame) and persistent data.
- Built a JSON and SQLite-backed persistence layer to store game state and user data.
- Implemented scheduling for input and event handling to keep the UI responsive during gameplay

Ray Tracing https://github.com/ChuzaWick420/rayTracing

- Developed a multi-threaded ray tracing renderer in C++ that distributes pixel computation across a configurable number of threads, boosting image generation depending upon hardware (4x on 6 cores processor) and shows the result on a SFML window.
- Built a JSON-driven configuration system covering four parameter groups - scene objects (size, color, material, position), camera (position, orientation, focal length), image output (dimensions, samples per pixel), and thread count - so scenes can be re-rendered without touching code.

Technical Writing https://github.com/ChuzaWick420/technical_writing

- Wrote an ATS friendly resume template with control parameters such as font family, font size, spacing, colors and sequence of modularized sections.
- Organized written samples (resumes, articles, cover letters etc) as skill polishing and proof.

Work Experience

Inbound Agency LTD Jan 2024 - Feb 2024
Frontend Developer Intern *Multan, Pakistan*

- Developed UI components using HTML, CSS, JavaScript, and React.
- Performed database queries and content updates through a Strapi headless CMS.

Education

Virtual University of Pakistan Oct 2022 - Sep 2026
Bachelor of Science, Computer Science *Lahore, Pakistan*

- Cumulative GPA: 3.7/4.0